

Project Design Phase

Solution Architecture

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Team ID: LTVIP2025TMID35065

Project Name: Trafficelligence: Advanced Traffic Volume Estimation with Machine Learning

Maximum Marks: 4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Trafficelligence Solution Architecture Overview:

The architecture of the Trafficelligence system consists of the following key components:

1. **Data Sources**: Historical traffic data, weather data, timestamp, holiday flags, and external events.
2. **Data Ingestion Layer**: Prepares, cleans, and formats the raw input data using ETL (Extract, Transform, Load) pipelines.
3. **Machine Learning Engine**:
 - Trains regression models (Linear Regression, Random Forest, XGBoost, SVR).
 - Evaluates model performance using metrics like R^2 Score, RMSE, and MAE.
 - Selects the best-performing model for deployment.
4. **Prediction Service**: Serves predictions in real-time or batch mode using a RESTful API.
5. **Dashboard/Frontend**: Displays real-time predicted traffic volumes to authorities for action and visualization.
6. **Monitoring & Logging**: Logs prediction data, errors, and model drifts for future tuning.

This architecture allows easy integration with city surveillance systems, traffic signals, or cloud-based smart city platforms.

Example - Solution Architecture Diagram:

Figure 1: Architecture and data flow of the Trafficelligence application (Insert your project diagram here).

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>